NBA All-Star Prediction



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01 Introduction



Our Question

Which NBA players are projected to make the 2026 All-Star roster?



Context

- All-Star selection depends on performance + popularity
  - Fans (50%), Players (25%), Media (25%)
- Combine traditional stats with sentiment analysis
- Dataset includes:
  - Player metrics: PPG, assists, rebounds, efficiency, usage rate
  - Sentiment scores from ESPN & online popularity from google trend



Project Motivation

- Stakeholders: NBA media teams, front offices, fantasy/betting platforms (PrizePicks, DraftKings)
- Benefits:
  - Early All-Star forecasts backed by data, not polls
  - Analytics-driven narratives for media
  - Marketability insights for teams
  - More accurate projection tools for fantasy & betting platforms



02

Datasets







Overview

1. Season-long performance stats (official NBA box score data)
2. Media sentiment scores (scraped ESPN news articles)
3. Popularity time-series (Google Trends interest over time)

Dataset 1: Season Performance Data

- Source: Scraped from **Basketball-Reference** (19–20, 20–21, 22–23, 23–24 seasons)
- Structure: One row per team stint (traded players appear multiple times)
- Columns: demographics + statistics (points, rebounds, assists, etc.)
- Cleaning:
  - Removed duplicate entries
  - Kept the latest team stint for each player

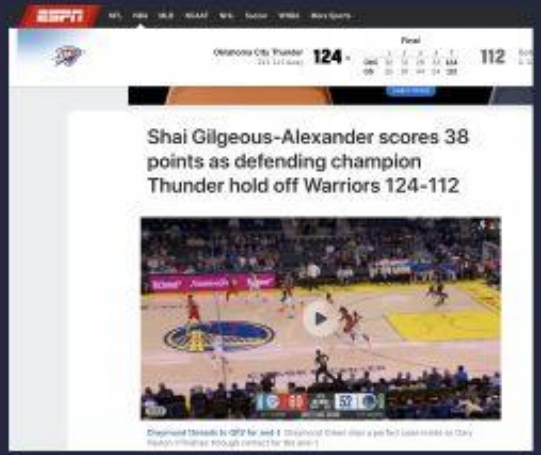


Per Game | Upgraded | Share & Export | When table is sorted, hide non-qualifiers for

Regular Season

Rk	Player	Age	Team	Pos	G	GS	MP	FG	FGA	FG%	3P	3PA	3P%	2P	2PA	2P%
1	Luka Dončić	26	LAL	PG	16	16	36.9	10.9	22.9	.478	3.7	10.8	.341	7.3	12.1	.601
2	Shai Gilgeous-Alexander	27	OKC	PG	21	21	33.5	10.7	19.7	.545	2.1	5.1	.411	8.6	14.6	.592
3	Tyrese Maxey	25	PHI	PG	19	19	**40.7**	10.7	**22.9**	.468	3.7	9.7	.384	7.0	13.2	.530
4	Giannis Antetokounmpo	31	MIL	PF	16	16	30.8	**11.8**	18.4	.639	0.6	1.4	.435	**11.1**	**16.9**	.657
5	Donovan Mitchell	29	CLE	SG	20	20	34.3	10.6	20.9	.508	4.1	10.4	.391	6.6	10.5	.624
6	Nikola Jokić	30	DEN	C	20	20	34.9	10.4	16.8	.621	2.1	4.8	.438	8.3	12.0	.695
7	Anthony Edwards	24	MIN	SG	16	16	33.3	9.8	19.9	.491	3.5	8.6	.409	6.3	11.3	.552
8	Jalen Brunson	29	NYK	PG	17	17	34.4	10.1	21.2	.478	2.8	7.5	.370	7.4	13.7	.538
9	Jaylen Brown	29	BOS	SF	20	20	33.1	10.6	21.6	.490	2.1	5.9	.347	8.5	15.7	.543
10	Cade Cunningham	24	DET	PG	18	18	36.8	10.0	21.9	.457	1.9	6.3	.301	8.1	15.6	.520
11	Austin Reaves	27	LAL	SG	17	17	36.6	8.6	16.6	.519	2.8	7.6	.364	5.9	9.1	.649
12	Lauri Markkanen	28	UTA	PF	20	20	35.6	9.4	20.0	.469	3.1	8.6	.360	6.3	11.4	.551
13	Stephen Curry	37	GSW	PG	16	16	31.1	9.1	19.4	.471	**4.7**	**12.0**	.391	4.4	7.4	.602
14	James Harden	36	LAC	PG	20	20	35.1	7.6	17.4	.435	3.6	9.8	.369	4.0	7.6	.520
15	Deni Avdija	25	POR	SF	20	20	34.1	8.0	16.8	.478	2.6	6.8	.378	5.5	10.1	.543
16	Kevin Durant	37	HOU	SF	16	16	35.9	8.5	17.4	.489	1.8	4.8	.377	6.7	12.6	.533

Dataset 2: Media Sentiment



- Scraped ESPN news articles
- Source: Google Custom Search API (espn.com/nba)
- Structure: One row per headline per player per season
- Raw Values:
  - ESPN article titles, Snippet text, Article count per player (totalResults), Google metadata (title, link, snippet)
- Cleaning:
  - Removed invalid/missing headlines, Deduplicated repeated titles, Extracted article dates using regex, Standardized player names, Dropped rows without valid text for sentiment analysis

Dataset 3: Online Popularity (Google Trend)

- Source: Google Trends
- Structure: One row per day, one column per player
  - Timeframe: start of season ~ final voting date
- Raw Values:
  - 0-100 value based on a relative popularity of a search term, scaled so the peak interest is 100.





03

Methods







Methods: A Two-Model Approach

Solution: Split into Two Models

- Model 1 (Performance): Objective side (Player Stats, Efficiency, Team Success).
- Model 2 (Sentiment/Popularity): Subjective side (Hype, Media Narratives, Fan Attention).

Why Our Approach Works:

- Pure performance misses underrated players; popularity alone overrated.
- Mirrors Voting: reflects Fan (50%), Media (25%), Player (25%) split.

Methods



Model 1: Statistical Performance Model

- Features: PTS, AST, REB, STL, BLK, FG%, 3P%, FT%, +/-, Win %, Usage Rate, GS, MIN
- Models used: logistic regression, neural network, random forest (ensemble)
- Output: performance-based All-Star probability (0–1)



Model 2: Sentiment & Popularity Model

- Inputs:
  - Media sentiment (ESPN headlines)
  - Google Trends popularity
- Techniques: sentiment scoring + time-series normalization
- Output: popularity-driven All-Star probability (0–1)

Final Step: Final score (Weighted Ensemble)

- Goal: Combine objective (Model 1: Stats) and subjective (Model 2: Hype) scores to align with the official voting structure (Fans, Media, Players).
- **Model Components:**
- Model 1: Performance-based (Ensemble of Logistic Regression, Neural Net, Random Forest).
- Model 2: Sentiment & Popularity (ESPN Sentiment Analysis + Google Trend).
- Weighted Ensemble Formula:
  - **Final Score = α * Model 1 + (1 - α) * [β * ESPN + (1 - β) * Google Trend]**
- Parameter Tuning (Grid Search):
  - α: Weighting between performance (stats) vs. hype (sentiment/popularity).
  - β: Relative weight between ESPN sentiment vs. Google popularity.Output: A single, continuous All-Star probability [0, 1] per player.

Model 1 Overview





- Goal: Predict NBA All-Star selections
- Dataset: Player stats (per-game + advanced metrics)
- Approach: Build 3 models → compare → ensemble

Model 1 Overview

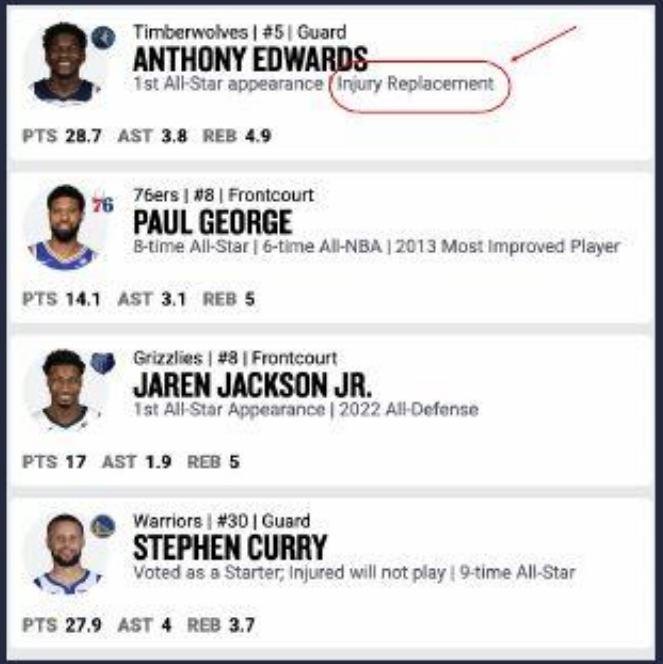
Data Preparation (Model 1)

Data Cleaning

- Missing data from players
  - Set to 0
- Ambiguous team names (ex: 2TM)
  - Remove player rows

Data Enrichment

- Adding All Star boolean column (target label)
  - Manually check on NBA website for all 5 years
  - Ignore injury replacement players



All Star
1
0
1
1
1
...
0
0
0
0
0

Data Preparation (Model 1)

```
all_star_list_2024 = ['Giannis Antetokounmpo', 'Bam Adebayo', 'Tyrese Haliburton', 'Damian Lillard', 'Jayson Tatum', 'Paolo Banchero',
                'Jaylen Brown', 'Jalen Brunson', 'Tyrese Maxey', 'Donovan Mitchell', 'Julius Randle', 'Joel Embiid',
                'LeBron James', 'Luka Dončić', 'Kevin Durant', 'Shai Gilgeous-Alexander', 'Nikola Jokić', 'Devin Booker', 'Stephen Curry', 'Anthony Davis',
                'Anthony Edwards', 'Paul George', 'Kawhi Leonard', 'Karl-Anthony Towns']

all_star_list_2023 = ['Giannis Antetokounmpo', 'Lauri Markkanen', 'Donovan Mitchell', 'Ja Morant', 'Jayson Tatum', 'Bam Adebayo',
                'Shai Gilgeous-Alexander', 'DeMar DeRozan', 'Jrue Holiday', 'Damian Lillard', 'Domantas Sabonis', 'Kevin Durant',
                'LeBron James', 'Luka Dončić', 'Joel Embiid', 'Kyrie Irving', 'Nikola Jokić', 'Jaylen Brown', 'Stephen Curry', 'Paul George',
                'Tyrese Haliburton', 'Jaren Jackson Jr.', 'Julius Randle', 'Zion Williamson']

all_star_list_2022 = ['Kevin Durant', 'Joel Embiid', 'Ja Morant', 'Jayson Tatum', 'Andrew Wiggins', 'Trae Young',
                'Devin Booker', 'Rudy Gobert', 'Zach LaVine', 'Khris Middleton', 'Karl-Anthony Towns', 'Draymond Green',
                'LeBron James', 'Luka Dončić', 'Giannis Antetokounmpo', 'Stephen Curry', 'Nikola Jokić', 'DeMar DeRozan', 'Jimmy Butler', 'Darius Garland',
                'James Harden', 'Donovan Mitchell', 'Chris Paul', 'Fred VanVleet']

all_star_list_2021 = ['Giannis Antetokounmpo', 'Stephen Curry', 'Luka Dončić', 'LeBron James', 'Nikola Jokić', 'Jaylen Brown',
                'Paul George', 'Rudy Gobert', 'Damian Lillard', 'Chris Paul', 'Ben Simmons',
                'Bradley Beal', 'Kyrie Irving', 'Kawhi Leonard', 'Jayson Tatum', 'Zion Williamson', 'James Harden', 'Zach LaVine', 'Donovan Mitchell',
                'Julius Randle', 'Nikola Vučević', 'Kevin Durant', 'Anthony Davis', 'Joel Embiid']

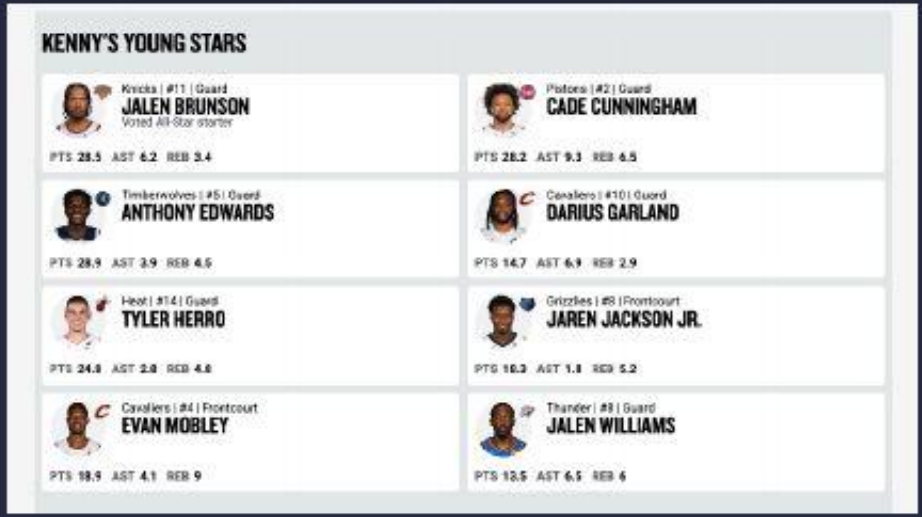
all_star_list_2020 = ['Anthony Davis', 'Kawhi Leonard', 'Luka Dončić', 'Damian Lillard', 'James Harden', 'Ben Simmons',
                'Nikola Jokić', 'Jayson Tatum', 'Chris Paul', 'Russell Westbrook', 'Domantas Sabonis', 'LeBron James',
                'Giannis Antetokounmpo', 'Joel Embiid', 'Pascal Siakam', 'Kemba Walker', 'Trae Young', 'Khris Middleton', 'Bam Adebayo', 'Rudy Gobert',
                'Jimmy Butler', 'Kyle Lowry', 'Brandon Ingram', 'Donovan Mitchell']
```

Data Preparation (Model 1)

Train and Test

- Train: 2020, 2021, 2022, 2023 (concat)
- Test: 2024
- Predict: 2026
- Skip 2025 due to different format

2024-2025 Data



The 3 Models



1. Logistic Regression

- Linear model for probability prediction
- Fast, interpretable baseline
- Outputs smooth probability scores
- Useful for identifying impactful features



Neural Network

- 2-layer fully connected network
- Learns complex nonlinear structures
- Produces probability output



3. Random Forest

- Ensemble method
- Averages results of individual trees
- Improve accuracy while controlling overfitting
- Learns complex nonlinear structures

Ensembling the Outputs

Comparing Model Outputs

- Each model outputs a probability:
  - P_log_reg
  - P_random_forest
  - p_neural_net
- Different models emphasize different statistics
- Variation across models helps reduce bias

Ensemble Method → Final Output

- Final prediction = average of three model probabilities
- Formula:

$$p_{\text{ensemble}} = \frac{p_{\text{LR}} + p_{\text{RF}} + p_{\text{NN}}}{3}$$

- Sort players b
- Set top-k players to 1 for All Star Prediction
  - Compare with actual label
- Produced top predicted All-Stars list
- More stable than any single model
- Ensemble performed best overall

Challenge (Model 1)

Drastic Class Imbalance

- ~500 players per season
- Only 24 all star players
  - ~5% of the target labels are 1 (All Star); rest are 0 (Non-All star)
- Accuracy is not a good error metric
- Solution:
  - Focus on precision, recall, f1



Initial Train Data Evaluation

```
Logistic Regression AUC: 0.9919
Random Forest AUC: 1.0000
Neural Network AUC: 1.0000
Ensemble AUC: 1.0000
Ensemble Accuracy: 0.9992
                   precision    recall  f1-score

Not All-Star (0)        1.00      1.00      1.00
    All-Star (1)        0.99      0.99      0.99

        accuracy                            1.00
       macro avg        0.99      0.99      0.99
    weighted avg        1.00      1.00      1.00
```

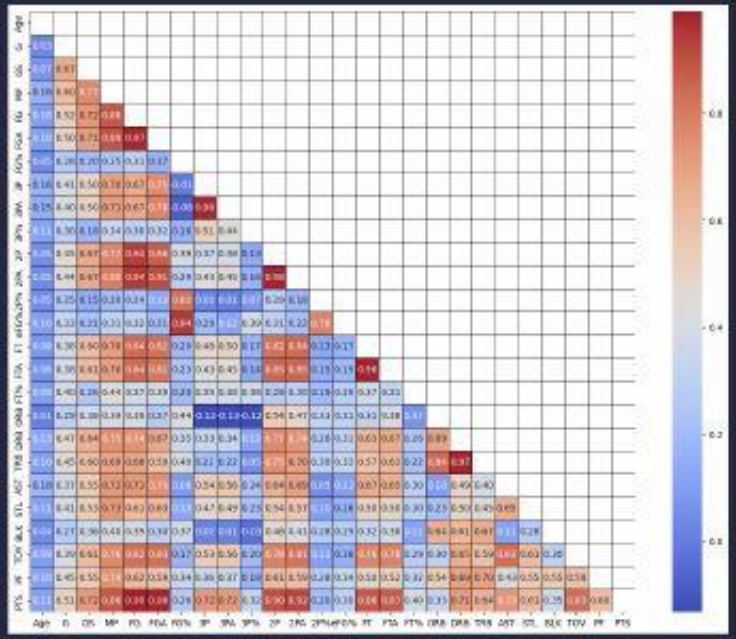
Challenge (Model 1)

Overfitting

- Reason 1: Many high correlated features
  - Ex:
    - FG vs PTS: 0.99
    - FGA vs PTS: 0.98



Feature Correlation Matrix



26 numerical features in raw data

Challenge (Model 1)

Overfitting

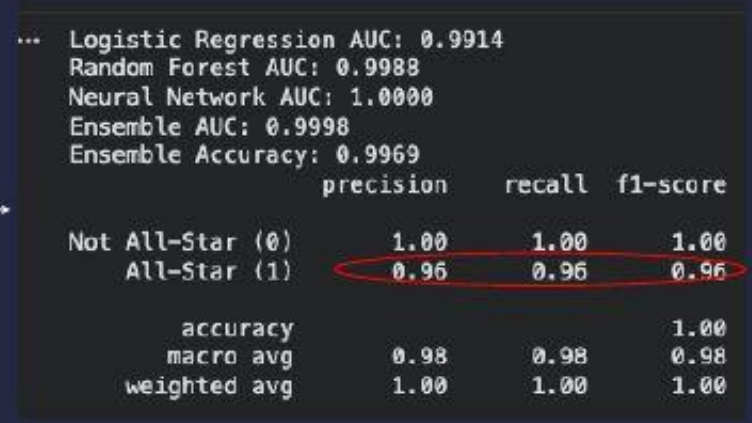
- Reason 2: Models failed to generalize
  - Random Forest & Neural Network
- Solutions:
  - Random Forest
    - Max_depth = None -> 5
    - Min_samples_split = 2 -> 5
  - Neural Network
    - Add alpha = 0.01
    - hidden_layer_sizes=(64, 32) -> (32, 16)

Train Data Evaluation

```
Logistic Regression AUC: 0.9919
Random Forest AUC: 1.0000
Neural Network AUC: 1.0000
Ensemble AUC: 1.0000
Ensemble Accuracy: 0.9992
                  precision    recall  f1-score

Not All-Star (0)       1.00      1.00      1.00
    All-Star (1)       0.99      0.99      0.99

        accuracy                           1.00
       macro avg       0.99      0.99      0.99
    weighted avg       1.00      1.00      1.00
```



```
Logistic Regression AUC: 0.9914
Random Forest AUC: 0.9988
Neural Network AUC: 1.0000
Ensemble AUC: 0.9998
Ensemble Accuracy: 0.9969
                  precision    recall  f1-score

Not All-Star (0)       1.00      1.00      1.00
    All-Star (1)       0.96      0.96      0.96

        accuracy                           1.00
       macro avg       0.98      0.98      0.98
    weighted avg       1.00      1.00      1.00
```

Test Data Evaluation (2024 All Star)

```
Ensemble Area Under the ROC Curve (AUC): 0.9918
Ensemble Accuracy: 0.9756
Ensemble Classification Report:
                  precision    recall  f1-score

Not All-Star (0)       0.99      0.99      0.99
    All-Star (1)       0.67      0.67      0.67

        accuracy                           0.98
       macro avg       0.83      0.83      0.83
    weighted avg       0.98      0.98      0.98
```



```
Ensemble Area Under the ROC Curve (AUC): 0.9931
Ensemble Accuracy: 0.9817
Ensemble Classification Report:
                  precision    recall  f1-score

Not All-Star (0)       0.99      0.99      0.99
    All-Star (1)       0.75      0.75      0.75

        accuracy                           0.98
       macro avg       0.87      0.87      0.87
    weighted avg       0.98      0.98      0.98
```

Model 2 Overview



Model 2 Overview

- Focuses on Sentiment / Popularity
  - Do people like the player?
  - Is the player famous?





Sentiment Analysis



ESPN Sentiment Analysis



- Media accounts for 25% of the total vote
- Scrape ESPN Articles through Google Search Custom API
- Scrape 10 articles per player / per season
    - Efficient query -> (takeaways OR analysis OR grades OR breakdown)
    - Calculate Avg Sentiment Score of 10 articles -> [0, 1] using ntlk
    - Popularity Component (# total num of ESPN articles returned)
    - Sentiment Component (Avg Sentiment Score of 10 articles)
- final_sentiment_score = ALPHA * popularity_component + (1 - ALPHA) * sentiment_component

Popularity Analysis

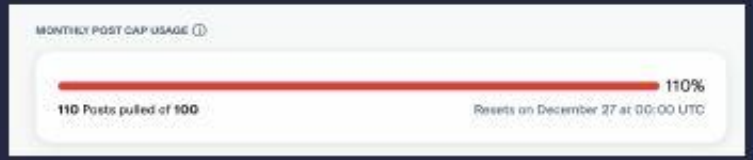
Google Trends

- Processing:
  - Players are processed in batches of 4 + LeBron James
  - Each batch is scaled relative to LeBron's maximum in the first batch
  - All data are normalized by the peak to a 0-100 scale.
- Data Extraction:
  - trends_mean= average search interest over time
  - trends_last30 = average search interest in last 30 days before selection
  - trends_max = highest value (peak) for a player
  - trends_slope = Slope of linear regression line, which indicates a player's trend direction and strength.
- Normalize and score: Each player's features are normalized, weights are defined, and combined to calculate a final score.

Challenges:
Insufficient Information:

- Reddit developer account rejected
- X (Twitter)
  - Unable scrape twitter post before 11/20, 2025
- Pytrends: Can only scrape 5 player's popularity each time







Results (Model 2)

Model 2: Sentiment Analysis

```
=== 2019–20 ===
Top-25 Precision: 0.480
Top-25 Recall:    0.480
Top-25 F1:        0.480
Top-25 IoU:       0.316
TP: 12, FP: 13, FN: 13

=== 2020–21 ===
Top-27 Precision: 0.704
Top-27 Recall:    0.704
Top-27 F1:        0.704
Top-27 IoU:       0.543
TP: 19, FP: 8, FN: 8

=== 2021–22 ===
Top-27 Precision: 0.593
Top-27 Recall:    0.593
Top-27 F1:        0.593
Top-27 IoU:       0.421
TP: 16, FP: 11, FN: 11
```

```
=== 2022–23 ===
Top-27 Precision: 0.556
Top-27 Recall:    0.556
Top-27 F1:        0.556
Top-27 IoU:       0.385
TP: 15, FP: 12, FN: 12

Training Precision: 0.58
Training Recall: 0.58
=== 2023–24 ===
Top-26 Precision: 0.615
Top-26 Recall:    0.615
Top-26 F1:        0.615
Top-26 IoU:       0.444
TP: 16, FP: 10, FN: 10

Testing Precision: 0.62
Testing Recall: 0.62
```

Model 2: Popularity Analysis

```
Season: 2019-2020
Top-26 Precision: 0.423
Top-26 Recall:    0.440
Top-26 F1:        0.431
Top-26 IoU:       0.275
TP: 11, FP: 15, FN: 14

Season: 2020-2021
Top-26 Precision: 0.538
Top-26 Recall:    0.519
Top-26 F1:        0.528
Top-26 IoU:       0.359
TP: 14, FP: 12, FN: 13

Season: 2021-2022
Top-26 Precision: 0.500
Top-26 Recall:    0.481
Top-26 F1:        0.491
Top-26 IoU:       0.325
TP: 13, FP: 13, FN: 14
```

```
Season: 2022-2023
Top-26 Precision: 0.500
Top-26 Recall:    0.481
Top-26 F1:        0.491
Top-26 IoU:       0.325
TP: 13, FP: 13, FN: 14

Season: 2023-2024
Top-26 Precision: 0.462
Top-26 Recall:    0.462
Top-26 F1:        0.462
Top-26 IoU:       0.300
TP: 12, FP: 14, FN: 14
```



04

Results & Conclusions



Base Model (Naive Model)

- **Imbalance data** (Each Season: 550+ players / 24 all star)
- Naive Model (Predict 24 players randomly)
  - Accuracy = 0.955
  - Precision = 0.045
  - Recall = 0.045
- Precision / Recall > Accuracy



Results (Combination of All Models)

Train

Best alpha: 0.78

Best beta: 0.000

Precision: 0.979

Recall: 0.979

```
...  Best alpha: 0.78000
     Best beta:  0.00000
     Best avg precision@k: 0.979
     Best avg recall@k:    0.979
```

Test

Precision: 0.73

Recall: 0.79



Alpha and Beta were determined through grid search

Model 1 dominates (alpha = 0.78), google trend more useful than ESPN sentiment

Hybrid score = 78% * model 1 + 22%[0% *ESPN + 100% *google trend]

Conclusion

Hybrid score = 78% * model 1 + 22%[0% *ESPN + 100% *google trend]

Key findings

- Model 1 were the most predictive of all star selection
  - Alpha = 0.78, giving heavy weight to on-court production
- Online Popularity contributed meaningful, outperforming media sentiment
  - Beta = 0 shows online popularity > ESPN sentiment
- Hybrid model achieves strong real world alignment
  - Precision 0.73, recall 0.79 on unseen test data

Model interpretation

- Driven primarily by performance with popularity providing the second strongest signal
- Sentiment (ESPN) had limited predictive power relative to other two
- The ensemble captures multiple dimensions of all stat voint : performance, media, and fan interest

Conclusion

Overall takeaway

- A hybrid model combining stat+popularity+sentiment provides a more complete and fair prediction than any individual model
- The method is flexible and can generalize to future seasons with updated data

Readiness for Use in Real-World

Model Prediction & Reliability

- Hybrid model achieves strong performance:
  Precision = 0.73, Recall = 0.79 on the test set
- Indicates the model captures real All-Star selection dynamics

Voting Aligned Prediction Capability

- Weighted ensemble reflects the real All-Star voting structure
- Stats (Model 1) → mirrors media/ player evaluation
- Popularity (Model 2) → mirrors fan voting Sentiment (Model 2) → complements narrative effect

Readiness for Use in Real-World

Practical Readiness

- Produces a single Hybrid All-Star Score per player
- Can generate real-world projections by ranking players across conferences & positions
- Easily extendable to live NBA data (updated box scores, trending metrics)

Real-World Impact

- Useful for analysts, fantasy players, and sports journalists
- Provides transparent, data-supported All-Star forecasts
- Reveals underlying selection biases (performance vs hype)

Potential Next Steps

Expand Data Resources

- Social media engagement metrics (Twitter/X, Reddit, TikTok)
- Add more nuanced sentiment signals (player mentions, tone shifts)

Improve Model Architecture

- Explore
  - Gradient-boosted ensembles (XGBoost, LightGBM)
  - transformer-based text models
- Add calibration to ensure predicted probabilities match real voting likelihood

Potential Next Steps

Interpretation & Bias Analysis

- Analyze mismatches between performance and fan popularity
- Identify underrated or overhyped players
- Study whether the model exposes systemic biases (market size, media coverage)



	Player
0	Nikola Jokić
1	Luka Dončić
2	Victor Wembanyama
3	James Harden
4	Shai Gilgeous-Alexander
5	Giannis Antetokounmpo
6	Tyrese Maxey
7	Donovan Mitchell
8	Austin Reaves
9	Stephen Curry
10	Kawhi Leonard
11	Jalen Johnson
12	Cade Cunningham
13	Miles Bridges
14	Paolo Banchero
15	Jaylen Brown
16	Anthony Edwards
17	Kevin Durant
18	Michael Porter Jr.
19	Deni Avdija
20	Josh Giddey
21	Tyler Herro
22	Julius Randle
23	Jalen Brunson

Results

West














East















THANK YOU
DUB NATION